

MOHAMMAD AHMADULLAH KHAN

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SUMMARY

ML Researcher (JMI & IIT Madras) specializing in **AI engineering**, low-latency systems, and open-source infrastructure. Skilled in optimizing ML models, building production RAG microservices (FastAPI, Docker, ChromaDB), and contributing to core libraries (Scikit-learn, Keras). Winner of **SIH 2024** and ranked **Top 0.3%** in the Amazon ML Challenge. Published **flwr-trees** to PyPI.

EDUCATION

Indian Institute of Technology Madras

Jan 2023 – Expected Dec 2026

Bachelor of Science in Data Science and Applications (SGPA: 7.67 / 10.0)

Chennai

– **Relevant Coursework:** Machine Learning Practice, Statistical Computing, Big Data Analytics, System Design.

Jamia Millia Islamia

July 2022 – June 2026

Bachelor of Technology in Computer Engineering (CGPA: 8.70 / 10.0)

Delhi

– **Relevant Coursework:** Responsive and Safe AI, Analysis of Algorithms, Data Mining, Database Management Systems.

Bal Bhawan School

2019 – 2021

Central Board of Secondary Education (CBSE)

Bhopal

– Class XII (PCM): 95.8% | Class X: 95.2% | **JEE Advanced Qualified**

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash

Machine Learning: PyTorch, Scikit-learn, TensorFlow, RAG, LLM Inference, OpenCV, Ollama, Local LLM Deployment

Tools & Systems: FastAPI, Flask, ChromaDB, Docker, Git (GitHub Actions), Linux, pytest, PySpark, Azure Databricks

Foundations: Data Structures & Algorithms, System Design, Linear Algebra, Optimization, Discrete Mathematics

ACHIEVEMENTS

- **Winner, Smart India Hackathon (SIH) 2024** | **NTRO:** Secured 1st place in Grand Finals (PS 1680) for developing a language-agnostic few-shot keyword spotting system.
- **Rank 98/32,000+ Teams, Amazon ML Challenge 2024** | Finished in the **Top 0.3% Nationwide** in a large-scale ML competition for product attribute extraction.
- **Winner, HackJMI 2024** | Awarded 1st place in the university-level flagship hackathon.
- **Qualified, National Standard Examination in Physics (NSEP) 2021** | Top 1% nationally.

EXPERIENCE

Machine Learning Intern

Jan 2026 – Present

Institute of Liver and Biliary Sciences (ILBS)

New Delhi

- **Clinical Data Engineering:** Architected a robust pipeline to merge and normalize heterogeneous longitudinal patient records from 10+ clinical sources for the **"Walk for Life"** multi-modal NCD detection project.
- **rPPG Predictive Modeling:** Developed a **contactless vital sign extraction model** using realtime webcam video feeds and rPPG signal processing to measure Heart Rate, HRV, and SpO2, actively researching computer vision techniques to correlate facial blood flow dynamics with blood pressure (BP).
- **Systems Integration and Optimization:** Built low-latency IPC mechanisms and data-validation constraints using Python and Linux tools to process concurrent telemetry feeds with transactional integrity.

OPEN SOURCE CONTRIBUTIONS

Scikit-learn, Keras Hub, Scikit-bio

Remote

Contributor (Open Source Ecosystem)

Jan 2026 – Present

- **Performance Engineering:** Optimized library efficiency via vectorization in FastICA (#33269) for a **24x speedup** in scikit-learn and reduced scikit-bio import latency by **37%** via lazy-loading (#2427).
- **Numerical Stability:** Resolved LiteRT export failures for masked architectures in **Keras Hub** (#2555).
- **Ecosystem Reliability:** Facilitated **Pandas 3.0** support in scikit-bio (#2425) and corrected feature-name validation logic across the scikit-learn preprocessing module (#33268).

PROJECTS

flwr-trees (Open-Source Federated Learning Library) | *Python, Flower, Scikit-learn, PyPI*

- **Architecture & Packaging:** Engineered an open-source library published to **PyPI** (`pip install flwr-trees`) that abstracts complex distributed networks into standard scikit-learn **BaseEstimator pipelines**, eliminating standalone client-server scripts while securing **100+ downloads**.
- **Memory Optimization:** Developed a custom **DiskTreeStore serialization layer** that spills fitted ensemble components to disk using lazy-loading iterators, enforcing a strict **O(1) memory floor** during large-scale model evaluation.
- **Network & Privacy Optimization:** Coded cyclic round-robin boosting for **XGBoost**, integrated **differential privacy** hooks, and designed a **histogram-bin exchange protocol** that reduces initial payload sizes by **99.8%**.

LUFY - Law Understandable For You | *FastAPI, RAG, ChromaDB, Sentence-Transformers, Docker*

- **Overview:** Built a full-stack legal document assistant run on local finetuned LLM that turns an uploaded contract into a plain language summary, a three tier risk analysis, and grounded Q&A, across English and **16 Indian languages**.
- **Service Architecture:** Built and containerized (**Docker**) a **FastAPI** microservice with lifespan-managed pre-warming of both embedding and reranker models, backed by a **31-test pytest suite**.
- **Hybrid Retrieval:** Fused dense (ChromaDB) and **BM25** lexical search via **Reciprocal Rank Fusion**, catching exact legal terms that similarity search alone misses.
- **Structure-Aware Chunking:** Detected **legal section headers** during chunking and tagged each chunk with its section, using query-to-section overlap as a retrieval boost.

RESEARCH EXPERIENCE

Few-Shot Language Agnostic Keyword Spotting | *TorchAudio, PyTorch, Transformers*

Oct 2025 – Dec 2025

- **Problem:** Investigated techniques to scale keyword spotting from 10 to **440 classes**, maintaining generalization despite extreme data scarcity in multilingual streams.
- **Engineering:** Implemented a **Transformer-encoder** with **Voice Activity Detection (VAD)** and MFCC extraction to optimize language-agnostic audio processing.
- **Outcome:** Successfully scaled to **44x more classes** with minimal accuracy degradation of **<10%** compared to fully-supervised baselines.

PUBLICATIONS

Premise Order Matters: Positional Bias in Nested Defeasible Reasoning in SLM and LLM

Aug 2026

M. A. Khan. Manuscript submitted to Association for Computational Linguistics Review 2026

Under Review

- **Analysis:** Evaluated order-invariance in 5 small LMs (3.8–9B parameters) using a 1230-instance benchmark, observing an accuracy drop of **13–23 percentage points** between standard and worst-case permutations during inference.
- **Finding:** Discovered a consistent **order-sensitivity asymmetry** across architectures, where largest failures occur when the meta-exception leads, suggesting models struggle with order-invariant hierarchical reasoning.
- **Metric:** Introduced the **Stepwise Transfer Score (STS)**, revealing that despite 80–90% final accuracy, correct predictions are supported by correct intermediate logic only **20–27%** of the time.

FIB-4 False Negatives in MASLD: Clinical Phenotypes and an Augmented Triage Mode

Sep 2026

Oral Presentation, APASL STC 2026 Kumamoto, Japan

Accepted

- **Premise:** FIB-4 (cutoff **<1.30**), the guideline first-line MASLD fibrosis screen, relies on AST, ALT, and platelet count, markers that stay near-normal in a recognised obesity-driven phenotype ($\text{BMI} \geq 30 \text{ kg/m}^2$, $\text{ALT/AST} < 40 \text{ U/L}$), leaving its accuracy for early-to-moderate (F2+) fibrosis poorly characterised.
- **Method & Result:** Applied gradient-boosted trees with **SHAP** across 40 clinical variables on NHANES 2017–2020 data ($N=1,998$) to show FIB-4 misses F2+ fibrosis in **16.6%** of low-risk-classified patients, rising to **37.7%** at $\text{BMI} \geq 40 \text{ kg/m}^2$, then built a supplementary model (FIB-4 Plus: FIB-4 + BMI + waist circumference + Hepatic Steatosis Index) that outperformed the NAFLD Fibrosis Score (**AUROC 0.837 vs. 0.749**, DeLong $p = 0.019$).
- Selected for **Oral Presentation** and awarded the **APASL International Travel Award**.

CERTIFICATIONS

Google Data Analytics Professional Certificate

Coursera

Google Career Certificates (SQL, R, Tableau)

NPTEL Certification: Responsible and Safe AI Systems

NPTEL

Funded by MoE, Govt. of India (Elite Certification)

LEADERSHIP & CO-CURRICULAR

- **General Secretary, IEEE JMI:** Led team of 15+ to execute 10+ technical workshops for **1,200+ students**.
- **Instructor, Intro to ML Workshop:** Designed and delivered a comprehensive ML curriculum to **150+ students**.
- **Chief Arbiter, Piper Chess Club:** Organized and adjudicated tournaments for **30+ participants**.